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Research Paper

Eucommia ulmoides Oliv. bark aqueous extract inhibits osteoarthritis in a rat model of osteoarthritis



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ABSTRACT

Ethnopharmacological relevance: *Eucommia ulmoides* Oliv. bark (EU) is a common traditional Chinese herbal medicine for treatment of osteoarthritis (OA), but its therapeutic effect on OA and the underlying mechanisms have not been fully clarified. Our previous study showed that *Eucommia ulmoides* Oliv. bark aqueous extract (EUE) had a protective effect on cartilage, and this study was aimed to investigate the anti-osteoarthritis effect and mechanisms of EUE in a rat model of osteoarthritis.

Materials and methods: Thirty-two 5-week-old specific pathogen-free Sprague–Dawley rats which were randomized into four even groups ($n=8$). Group A received sham operation while the OA model was established using the modified Hulth technique in groups B, C and D. For eight weeks after operation, in addition to routine feeding, group A received gavage with deionized water, group B with deionized water, group C with 1.35 g/kg/day EUE, and group D with 2.7 g/kg/day EUE. Eight weeks postoperatively, all of the animals were euthanized for radiological, gross and histopathological observations to evaluate the effect of EUE on OA and to determine its potential mechanisms.

Results: Radiological and histopathological observations showed that the articular degenerative changes were significantly more alleviated in groups C and D than in group B, while there were no obviously degenerative manifestations in group A. Mankin's scores in groups C and D were significantly lower than in group B ($P < 0.01$). The severity of OA was significantly less in group D than in group C ($P < 0.01$). The IL-1 β and IL-6 contents in serum and MMP-3 secretion in articular cartilage were significantly lower in groups C and D than those in group B ($P < 0.01$), and significantly lower in group D than those in group C ($P < 0.01$). Compared with group B, phosphorylated Akt was significantly down-regulated in groups C and D.

Conclusions: EUE may inhibit the progression of osteoarthritis by inhibiting the PI3K/Akt pathway to delay cartilage degeneration, reduce inflammatory cytokines and prevent MMP-3 secretion. Therefore, EU is a potential therapeutic agent for OA, but its efficacy is limited.

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Abbreviations: AAOS, American Academy of Orthopaedic Surgeons; ELISA, enzyme-linked immunosorbent assay kits; ECM, extracellular matrix; E U, *Eucommia ulmoides* Oliv. bark; EUE, *Eucommia ulmoides* Oliv. bark aqueous extract; HPLC, high-performance liquid chromatography; H & E, Hematoxylin and Eosin; IL-1 β , interleukin-1 β ; IL-6, interleukin-6; MMPs, matrix metalloproteinases; MMP-3, matrix metalloproteinase-3; NSAIDs, nonsteroidal anti-inflammatory drugs; OA, osteoarthritis; PI3K, phosphoinositide 3-kinase; TCM, traditional Chinese medicine

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1. Introduction

Osteoarthritis (OA) is a common arthritic disease and a major cause of disability in adult population. It was generally considered a whole joint disease characterized by cartilage destruction, subchondral bone sclerosis, and osteophyte formation, joint synovitis, and synovial angiogenesis (Suri and Walsh, 2012; Zhen et al., 2013; Zhang et al., 2014b). Moreover, cartilage degeneration was caused by an imbalance between anabolic and catabolic factors. Matrix metalloproteinases (MMPs) play an important role in the degenerative process of articular cartilage and may degenerate extracellular matrix (ECM) of articular cartilage, which mainly consists of collagen- α and aggrecan (Martel-Pelletier et al., 2001). Likewise, some pro-inflammatory cytokines like interleukin (IL)-1 β and IL-6 might

provoke cartilage degradation through up-regulating MMPs secretion (Sakao et al., 2009; Stannus et al., 2010; Santangelo et al., 2012). It was also reported that PI3K/Akt pathway is involved in the progression of OA and inhibition of this pathway decreases MMPs production by chondrocytes (Chen et al., 2013). However, the etiology and mechanisms of OA are still not very clear to date and there are no specific drugs especially effective for OA either. Currently, nonsteroidal anti-inflammatory drugs (NSAIDs) are commonly used for OA and recommended by American Academy of Orthopaedic Surgeons (AAOS) (Jevsevar et al., 2013). However, they are associated with serious adverse events, such as gastrointestinal (Ishijima et al., 2014) and cerebrovascular diseases (Lee et al., 2007). Therefore, a safe and effective anti-osteoarthritis agent is urgently demanded. Traditional Chinese herbal medicines which contain a large number of active compounds could provide a choice for this kind of agent (Ho et al., 2013; Tong et al., 2014).

Eucommia ulmoides Oliv. bark (EU), also known as *Du Zhong*, is traditionally used as a tonic medicine in China, Japan, Korea and other countries (He et al., 2014). Based on theories of traditional Chinese medicine (TCM), EU is used to tonify the liver and kidneys, strengthen tendons and bones, and prevent miscarriage. Pharmacological studies have indicated that the extract of EU exhibits effects of fortifying the muscles and lungs, lowering blood pressure, preventing miscarriage and improving bone biomechanical quality (Lee et al., 2005; Zhang et al., 2009, 2014a). Since it is an indispensable ingredient of many classic prescriptions, such as Bushen Huoxue decoction and Duhuo Jisheng decoction, the aqueous extract of EU has been widely used in traditional Chinese medicine for the treatment of hypertension, osteoporosis and OA (Anon, 2010). Of note, EU is the most frequently prescribed herb and Duhuo Jisheng decoction is the most commonly prescribed Chinese formula for OA (Chen et al., 2014). Our previous study demonstrated that *Eucommia ulmoides* Oliv. bark aqueous extract (EUE) has a protective effect on the articular cartilage in rats (Lu et al., 2013), but it was unknown whether the aqueous extract of the single herb had the anti-osteoarthritis effect in OA.

Therefore, the aim of the present study was to investigate the effect of EUE for the treatment of OA in a rat model and its potential mechanisms.

2. Materials and methods

2.1. Plant materials and aqueous extract

EU was purchased from Zhixin Pharmaceutical Co., Ltd. (Guangzhou, China) and authenticated macroscopically and microscopically according to the Chinese Pharmacopoeia (Anon, 2010). A 135 g sample of EU was extracted with water (1000 ml) at 100 °C for 60 min and then centrifuged at 3000 rpm for 20 min. The extraction was repeated thrice. The extracts were then combined and filtered. The filtrates were collected, concentrated with a vacuum evaporator until the volume was 500 ml and then stored at –20 °C until use.

2.2. High performance liquid chromatography analysis

The qualitative chemical profile (fingerprint) of EUE was analyzed by high-performance liquid chromatography (HPLC) as described in the Chinese Pharmacopoeia. HPLC was carried on a Waters Alliance (Waters Corp, American) e2695 HPLC system consisting of a 2998 photodiode array detector, gradient pump, autosampler and C18 column (150 mm × 4.6 mm, 5 μm). Solution A comprising acetonitrile–0.1% H₃PO₄ (5/95, v/v) and solution B (100% acetonitrile) were used as the mobile phase with the gradient elution. The gradient program was accomplished at 25 °C by a linear gradient elution from 6% to 28% acetonitrile for 0–35 min, 28% to

35% acetonitrile for 35–60 min and 35% acetonitrile for 60–70 min at a flow rate of 0.8 ml/min. The scanned wavelength was from 210 nm to 360 nm, and the detection wavelength was 230 nm. According to the principle of HPLC, the compounds were identified by individual peak retention times compared with the standard substances.

2.3. Animal environment and osteoarthritis model

Thirty-two specific pathogen-free Sprague–Dawley rats (5 weeks old, 200–220 g in weight) were purchased from the Animal Laboratory Center, Guangzhou University of Traditional Chinese Medicine [SCXK (Yue) 2008-0020]. All rats were housed in a specific pathogen-free environment, which was temperature-controlled (24 ± 1 °C) and humidity-controlled (30–40%), in a 12 h light and 12 h dark cycle (lights on at 6:00 a.m.). Feed and water were provided ad libitum. All animals were disposed according to the “Guide for the Care and Use of Laboratory Animals” (National Research Council Committee for the Update of the Guide for the Care and Use of Laboratory Animals, 2011). This study was approved by the Animal Care and Use Committee at Guangzhou University of Traditional Chinese Medicine (2008C067).

The rat models of OA were established in 24 randomized rats by performing unilateral knee joint surgery according to the modified Hulth technique (Hulth et al., 1970; Hayami et al., 2003). The remaining 8 rats that received sham operation served as blank controls (group A). Specific methods were as follows: the right knee joint cavity of rat was opened via a medial parapatellar incision under general anesthesia with 10% chloral hydrate (3 ml/kg) before the anterior cruciate ligaments were transected to remove the medial meniscus. All animals were permitted to run for 1 h every day after operation.

2.4. EUE administration

The 24 rat models of OA were randomly and evenly divided into groups B, C and D (n=8). For eight weeks after operation, in addition to routine feeding, no EUE but deionized water was administered by gavage in the rats in groups A and B, but 1.35 g/kg/day EUE was administered by gavage in the rats in group C and 2.70 g/kg/day EUE in group D.

2.5. Radiographic evaluation, gross observation and histopathologic analysis

Radiographic evaluation was performed in all rats after anesthesia with 10% chloral hydrate at the end of 8 weeks postoperatively. Digital images of the knee joint were taken using an Axiom Multix M radiographic unit (Siemens, Germany). The right knee joint cavity was opened and soft tissues around the knee were removed after the rats were euthanatized by cervical dislocation following general anesthesia. The surface of articular cartilage was observed. Macroscopic lesions were graded according to the method described previously (Lavery et al., 2010).

The distal femur was fixed with 10% neutral buffered formalin, followed by decalcification in 4% EDTA for histological evaluation after macroscopic observation. The decalcified samples were embedded in paraffin blocks and serial sections (4 μm thick). At last, the sections were stained with Hematoxylin and Eosin (H & E). Mankin's score was used to further evaluate the cartilage damage. Semi-quantitative histopathological grading was performed according to the modified Mankin's scores system for the degradation of articular cartilage (Mankin et al., 1971; Hayami et al., 2006), covering structure, chondrocyte number, chondrocyte clustering, proteoglycan content, and subchondral bone plate and/or tidemark change.

2.6. Measurements of IL-1 β and IL-6 in serum

Blood was sampled from the eyeball vein of rats before sacrificing, and centrifuged at 3000 rpm for 15 min. The serum was collected and tested for IL-1 β and IL-6 contents using commercial enzyme-linked immunosorbent assay kits (ELISA) (R & D Systems Inc., USA) according to the manufacturer's instructions. Optical density was detected at 450 nm through a SpectraMax M5 autoreader (Molecular Devices, USA), and the outcomes were expressed as pg/ml.

2.7. Measurement of MMP-3 in articular cartilage

Immunohistological analysis was performed as described in the manufacturer's instructions (Boster Biotechnology Co., Ltd, Wuhan, China). All sections were immunostained with the SABC immunohistochemistry method, respectively. Specific process was as follows: dewaxing and antigen retrieval were conducted in the articular sections of the distal femur, followed by incubation overnight at 4 °C with primary antibody MMP-3 (1:100), then incubation with the secondary antibody, and finally counterstain with diaminobenzidine (DAB) and haematoxylin. The results were assessed semi-quantitatively according to the percentage of positive cells in the randomly selected fields in each section. The positive ratio was evaluated by two experienced observers in a blinded manner. Dark brown cells were deemed to be positive.

2.8. Western-blotting analysis

Articular cartilage of the distal femur was harvested and stored at –80 °C before assay. The samples were separately ground under liquid nitrogen (Siebelt et al., 2013), then mixed with the lysis buffer 10 μ l and centrifuged at 3000 rpm for 20 min at 4 °C. Signal quantification was

performed as previously described (Zushi et al., 2009). Proteins were admixed with the Laemmli buffer and boiled for 5 min. The proteins were transferred onto nitrocellulose membranes which were then incubated overnight at 4 °C with the primary antibodies against Akt, p-Akt (Cell Signaling Technology, Inc., USA), and incubated with the secondary antibody (anti-mouse IgG) (Santa Cruz Biotechnology Inc., USA) for 2 h at room temperature. The blots were developed using a chemiluminescence reagent, with the films being exposed and then visualized with an Odyssey Two-Color Infrared Imaging System (LI-COR, USA). Protein bands were quantified by densitometric analysis using Photoshop CS4 (Adobe Systems Incorporated, California, USA).

2.9. Statistical analysis

The data, expressed as mean $\pm$ standard deviation (S.D.), were processed by the SPSS 20.0 software package and statistically analyzed using the one-way analysis of variance (ANOVA). Duncan's multiple range tests were performed to determine significant differences among the groups. Statistical differences were defined as P value $\leq$ 0.05.

3. Results

3.1. HPLC analysis of EUE and chemical profile

The HPLC analysis of EUE (Fig. 1) showed that over 11 peaks were detected within 80 min, among which 7 compounds were identified or deducted by comparing individual peak retention times with those of the authentic reference substances. They were (1) aucubin, (2) geniposidic acid, (3) chlorogenic acid, (4) geniposide, (5) pinosresinol diglucoside, (6) rutin, and (7) quercetin.

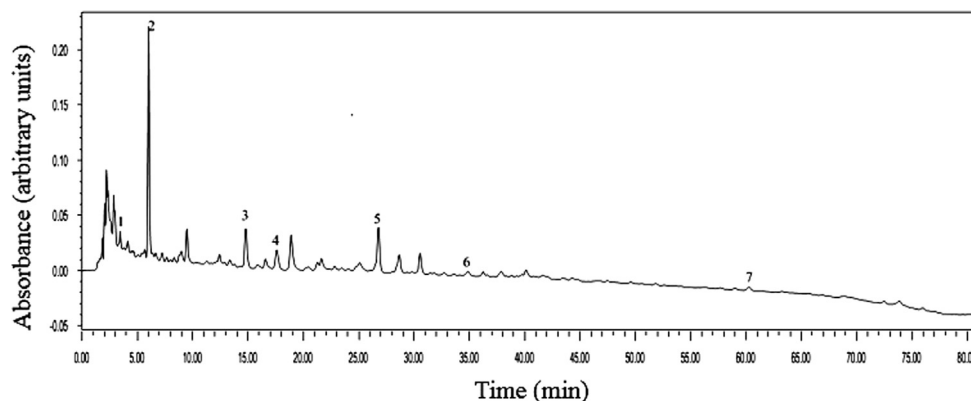


Fig. 1. HPLC chromatogram of EUE (1 mg/ml). Seven compounds were deducted by comparing individual peak retention times with those of the authentic reference substances. The peaks indicate the following: (1) C₁₅H₂₂O₉, (2) C₁₆H₂₂O₁₀, (3) C₁₆H₁₈O₉, (4) C₁₇H₂₄O₁₀, (5) C₃₂H₄₂O₁₆, (6) C₂₇H₃₀O₁₆, and (7) C₁₅H₁₀O₇.

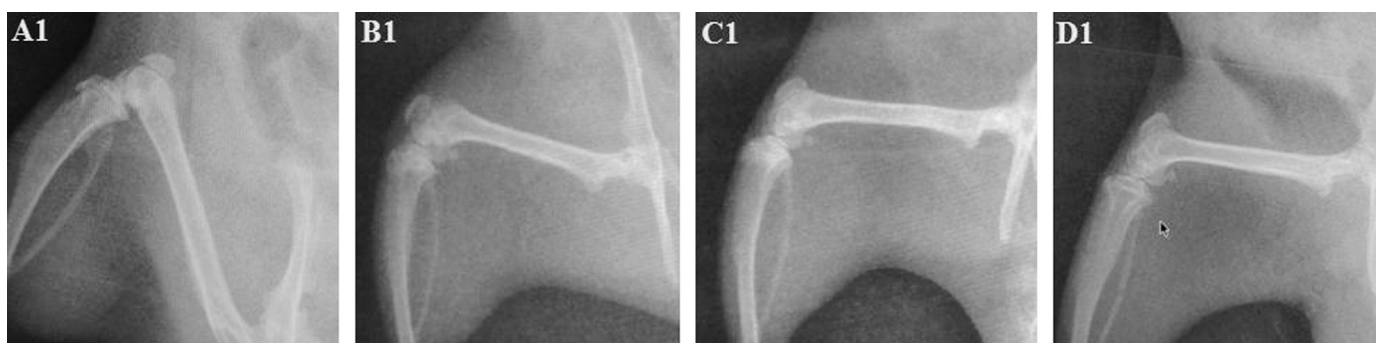


Fig. 2. Radiographic films of the rat knee joint in all the groups. Radiograph A1 from group A showing smooth articular surface; radiograph B1 from group B showing osteophyte, incomplete and thickening articular surface with sclerosis and deformation; radiograph C1 from group C showing little osteophyte and thickening articular surface with sclerosis and deformation; and radiograph D1 from group D showing little osteophyte and relatively smooth articular surface.

3.2. Effect of EUE at radiographic analysis in the OA rat model

Radiographic examination revealed obvious differences in the degenerative characteristics of the right knee joint among groups A, B, C, and D (Fig. 2). Osteophyte, incomplete and thickening articular surface with sclerosis and deformation were evident in the OA rat models (groups B, C and D) while smooth articular surface without sclerosis or osteophyte was observed in group A. The severity of degeneration was obviously decreased in a sequence from group B to group C to group D.

3.3. Effect of EUE at gross observation in the OA rat model

Gross observation revealed smooth and lustrous articular surface at the right knee joint in group A, but serious yellowish-white, lackluster cracks and cartilage defects in group B. As radiographic examination did, gross observation also revealed the severity of degeneration was obviously decreased in a sequence from group B to group C to group D (Fig. 3).

3.4. Effect of EUE on histopathologic evaluation in the OA rat model

H & E analysis showed damaged cartilage, disorganized chondrocyte clusters and rough cartilage surface in the OA rat models (groups B, C and D) but smooth surface and regular chondrocytes in group A. Consistent with radiographic and gross observations, H & E analysis also showed that the severity of cartilage damage was markedly lower in a sequence from group B to group C to group D (Fig. 4). Likewise, Mankin's scores were significantly reduced from group B to group C to group D ($P < 0.05$) (Table 1).

All the above results indicated that EUE prevented cartilage damage and reduced cellular abnormality.

3.5. EUE reduces serum IL-1 β and IL-6 levels

The values of IL-1 β and IL-6 were significantly higher in groups B, C, and D than those in group A ($P < 0.05$), and significantly decreased in a sequence from group B to group C to group D ($P < 0.05$), indicating that EUE inhibited serum IL-1 β and IL-6 secretion in the OA rat models (Fig. 5).

3.6. EUE inhibits MMP-3 secretion in articular cartilage

Immunohistochemistry examination showed that group B had a significant higher percentage of MMP-3 expression in chondrocytes compared with those of groups C and D ($P < 0.01$). The percentage of MMP-3 expression in chondrocytes was significantly different between groups C and D ($P < 0.01$). Group A had no MMP-3 expression (Fig. 6, Table 2).

3.7. EUE down-regulates p-Akt expression in articular cartilage

Western blot analysis revealed that p-Akt expression levels which were quantified by densitometric analysis were significantly down-regulated in groups C and D compared with that of group B, while it

Table 1
Mankin's scores of the four groups.

Groups	Mankin's scores
Group A	1.88 $\pm$ 0.83*
Group B	18.63 $\pm$ 4.51
Group C	3.88 $\pm$ 1.25*
Group D	2.63 $\pm$ 1.06* [#]

* $P < 0.05$ compared to group B.

[#] $P < 0.05$ compared to group C.

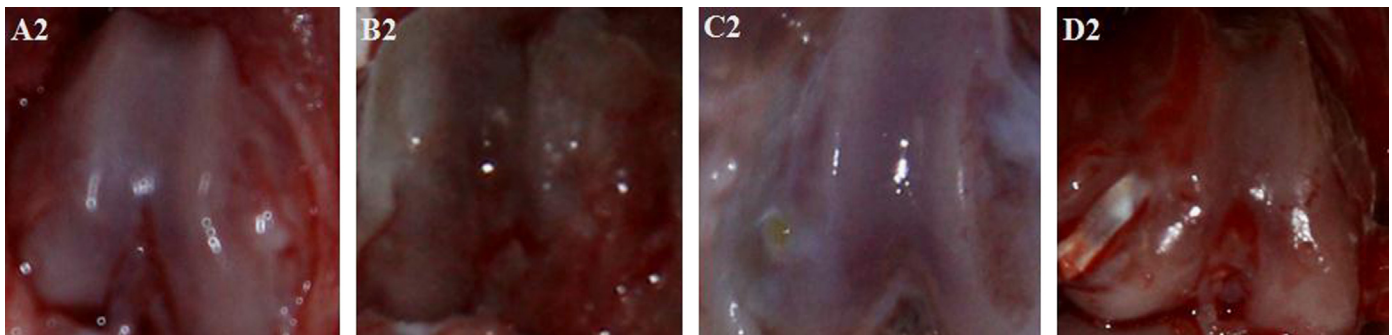


Fig. 3. Gross observations of the rat knee joint in all groups. Picture A2 from group A showing smooth and lustrous articular surface; picture B2 from group B showing yellowish-white, lackluster cracks and cartilage defects; picture C2 from group C showing little lackluster cracks and cartilage defects; and picture D2 from group D showing little cartilage degeneration.

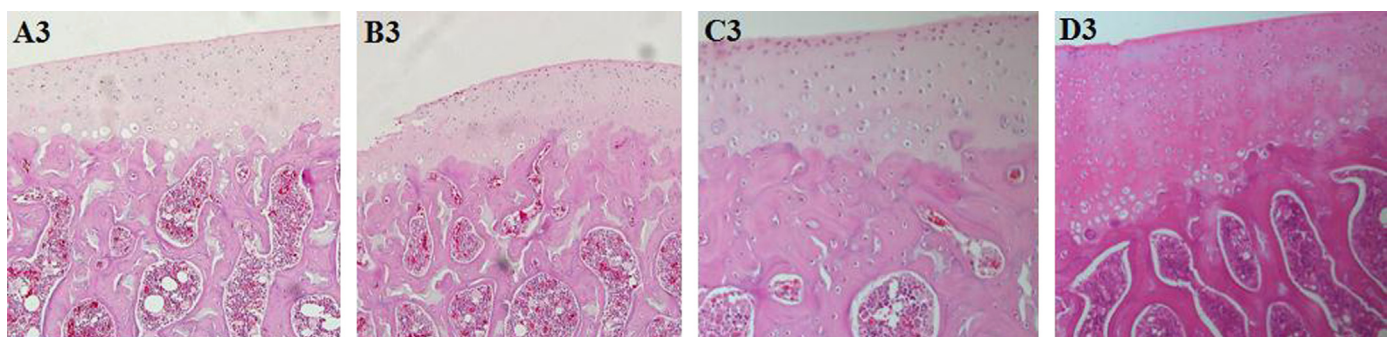


Fig. 4. Histopathologic profiles of rat knee joints in all groups. Micrograph A3 from group A showing smooth surface and regular chondrocytes; micrograph B3 from group B showing matrix discontinuity at the superficial zone and disorganized chondrocyte clusters; micrograph C3 from group C showing superficial fibrillation and disorganized chondrocyte clusters; and micrograph D3 from group D showing only superficial fibrillation.

was significantly up-regulated in group B compared with that of group A. The decrease in p-Akt expression level in group D was significantly greater than that in group C. There were no significant differences among groups A, B, C and D in Akt expression (Fig. 7).

4. Discussion

Our radiographic, gross and histopathologic observations qualitatively and semi-quantitatively determined that EUE inhibited the progression of OA by preventing matrix production and chondrocyte apoptosis in the OA rat models. Moreover, this study also revealed that EUE reduced secretion of IL-1 β and IL-6 to inhibit inflammatory reaction, prevented MMP-3 production to protect articular cartilage, and down-regulated p-Akt protein expression.

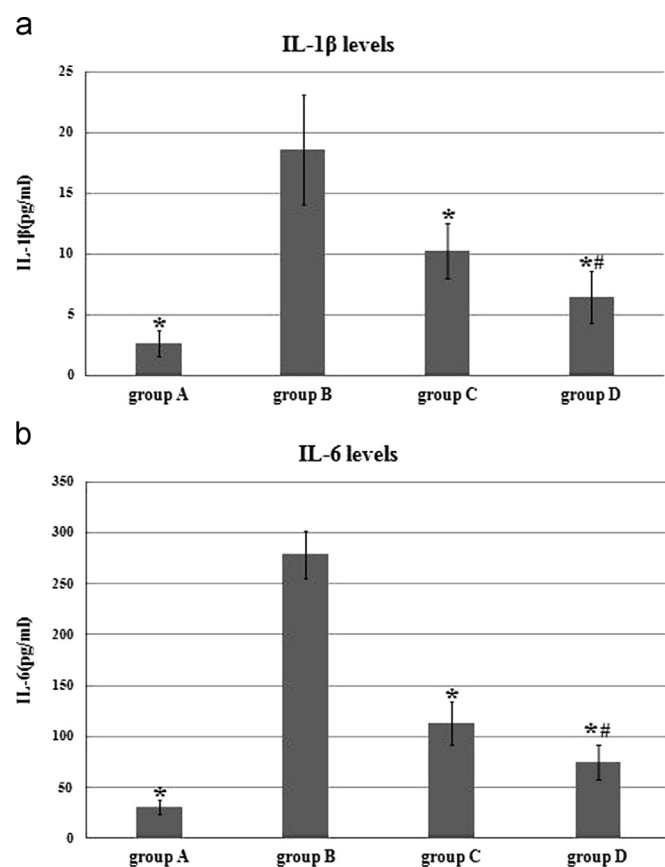


Fig. 5. Serum IL-1 β and IL-6 levels in all the groups. * $P < 0.05$ compared to group B, # $P < 0.05$ compared to group C.

EU contains many active ingredients that may be used to authenticate this Chinese herbal medication, of which aucubin, geniposidic acid, chlorogenic acid, geniposide, pinosresinol diglucoside, rutin, and quercetin are the chief ones according to the Chinese Pharmacopoeia. Our test confirmed that the EU used for this study was authentic.

EU is a common traditional Chinese herbal medicine that has a long history of use for OA in China. To date, it is still an extremely effective herbal drug for therapy of OA in modern traditional Chinese medicine. Its usage and safety have been clinically identified (Li et al., 2000; Wang et al., 2005). Our previous study found that it decreased the contents of MMP-1, MMP-3 and MMP-13 in synovial fluid in the OA rat models, indicating that it might protect articular cartilage (Lu et al., 2013). However, its effect on OA and the underlying mechanisms are not clear. Therefore, the present study was aimed to systematically evaluate the effects of EUE on OA and its potential mechanisms.

Our radiological results showed that EUE decreased the severity of articular degeneration in a dose-effect manner but osteophytes and subchondral sclerosis were still observed in radiographic images, suggesting that EU could not completely inhibit the OA progression. At the same time, EUE also prevented pathological damage to articular cartilage, but partial chondrocyte necrobiosis was observed in the rats treated with EUE. Likewise, although we found that EUE prevented extracellular matrix secretion and chondrocyte apoptosis, its effects on preventing articular degeneration were not thorough enough. All these results indicate that EU may have a positive effect on OA but the efficacy of this herbal drug is limited.

As we all know, the pro-inflammatory cytokines like IL-1 β and IL-6 play an important role in the progression of OA. Both IL-1 β and IL-6 are deemed to display catabolic properties that influence the pathophysiological processes of OA (Rutgers et al., 2010; Yu and Kim, 2013). Previous studies showed that the contents of IL-1 β and IL-6 are higher in the serum of OA patients compared with those of the control group (Hussein et al., 2008; Lee et al., 2011), and closely associated with OA patients' pain as well. This study found that the contents of IL-1 β and IL-6 were significantly higher

Table 2
MMP-3 expressions of the four groups.

Groups	MMP-3 (%)
Group A	3.63 $\pm$ 1.85*
Group B	44.25 $\pm$ 9.13
Group C	20.63 $\pm$ 6.55*
Group D	16.25 $\pm$ 5.34*#

* $P < 0.05$ compared to group B.

$P < 0.05$ compared to group C.

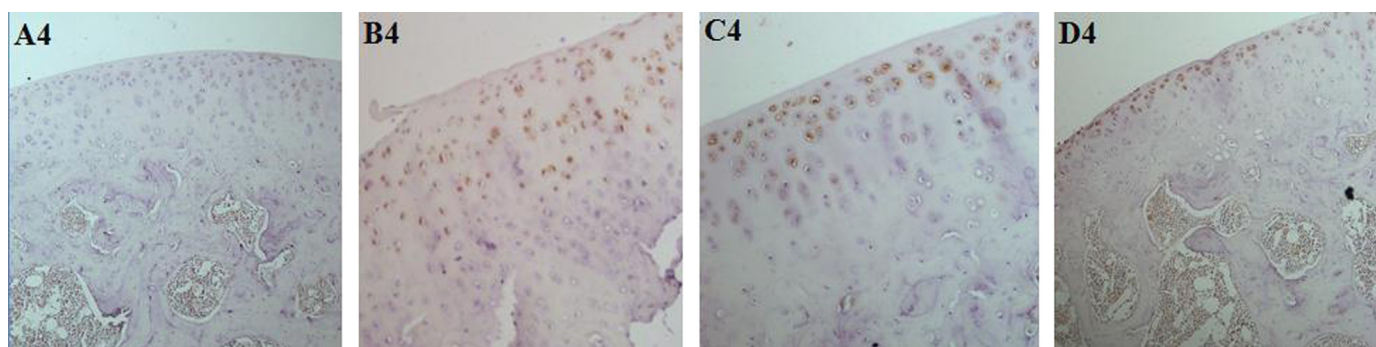


Fig. 6. Distribution of chondrocyte matrix metalloproteinase-3 (MMP-3) staining of articular cartilage in all groups. Micrograph A4 from group A showing no MMP-3 staining; micrograph B4 from group B showing a high percentage of MMP-3 staining in distal femoral articular cartilage chondrocytes; micrograph C4 from group C showing middle percentage of MMP-3 staining; and micrograph D4 from group D showing low percentage of MMP-3 staining.

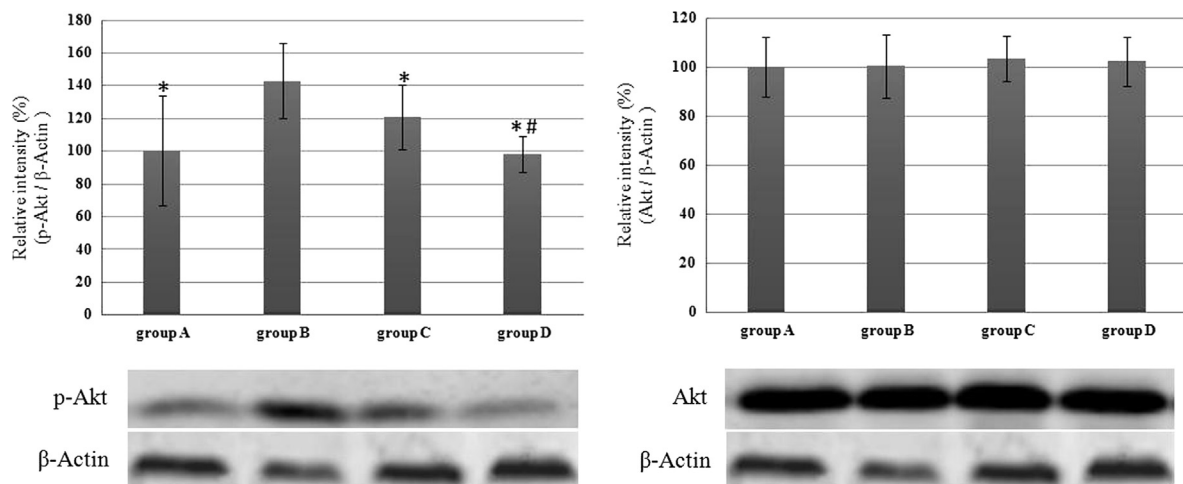


Fig. 7. The protein expression of p-Akt and Akt in all groups. Expression levels of Akt and p-Akt were assessed by Western blotting, and band intensities were quantified by densitometric analysis. Data are presented as mean $\pm$ S.E.M. * $P < 0.05$ compared to group B, # $P < 0.05$ compared to group C.

in the serum of OA rats but intervention of EUE significantly decreased them in a dose-effect dependent manner. Similarly, previous study revealed that EU resisted inflammatory reaction by inhibiting TNF- α and IL-6 production and by suppressing secretion of cyclooxygenase-2 and prostaglandin E_2 in the LPS-induced mouse peritoneal macrophages (Kim et al., 2012). All these suggest that EUE may inhibit inflammatory reaction to partially prevent and slow the progression of OA.

OA is characterized by degenerative alteration in joints, mainly included chondrocyte apoptosis and extracellular matrix (ECM) degradation. MMPs are considered as the main proteolytic enzyme which may degrade ECM. MMP-3, produced by chondrocytes, plays an important role in the development of cartilage damage (Bondeson et al., 2006). This study found that EUE decreased MMP-3 production in articular cartilage. This may be a mechanism for the effect of EU on deferring degenerative progression of articular cartilage.

Previous study indicated that the PI3K/Akt signaling pathway was involved in the regulation of chondrocyte apoptosis and ECM degradation and that its activation promoted MMPs production in chondrocytes (Chen et al., 2013). We found that EUE inhibited phosphorylation of Akt protein expression to block the PI3K/Akt signaling pathway. Previous study also showed that EUE significantly suppressed 6-OHDA-induced phosphorylation of Akt in SH-SY5Y cells (Kwon et al., 2014), similar to our results. Therefore, we conclude that EUE may prevent the PI3K/Akt signaling pathway to resist inflammatory reaction and defer OA. It is reasonable to inhibit the PI3K/Akt signaling pathway as a possible therapy of OA.

5. Conclusions

In summary, this study provides evidence on the anti-osteoarthritis effects of EUE and finds that EU may prevent the progression of OA by reducing pro-inflammatory cytokines and MMP-3 production and inhibiting the PI3K/Akt signaling pathway. This study also finds that the anti-osteoarthritis effects of EUE are limited. These findings suggest that EU may be an effective ancillary agent for the prevention and therapy of OA.

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Corrigendum

Corrigendum to 'Eucommia ulmoides Oliv. bark aqueous extract inhibits osteoarthritis in a rat model of osteoarthritis'



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The authors regret a mistake in Fig. 7

The figures of western blot for Akt and β -Actin were mislabeled.

The correct form was listed below.

The authors would like to apologize for any inconvenience caused.

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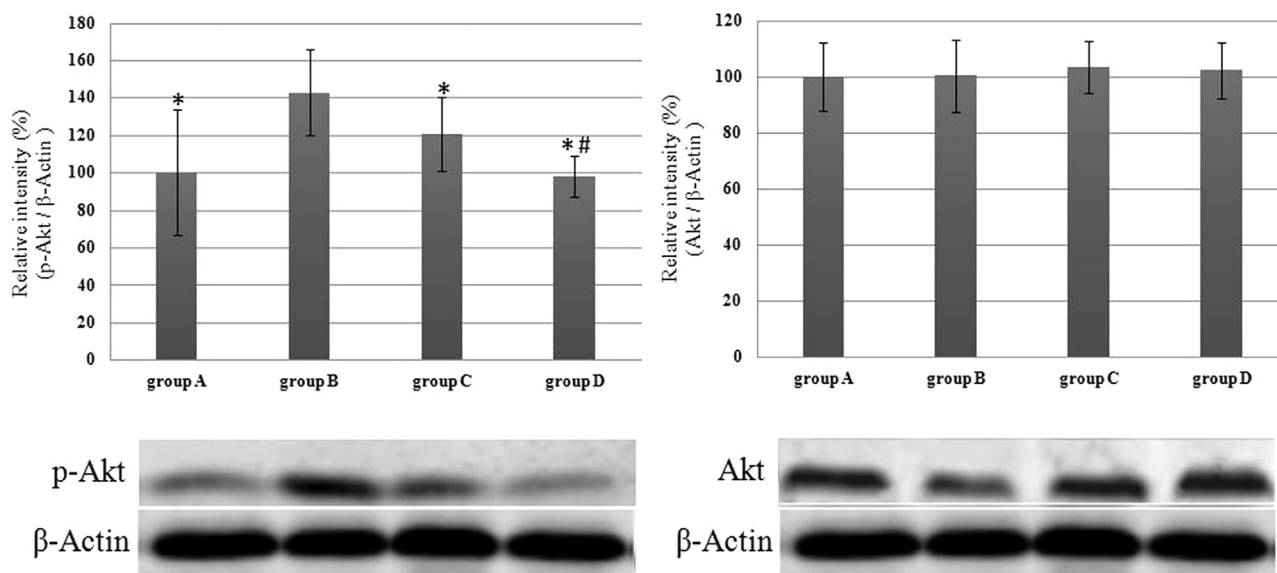


Fig. 7

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